



Attaining robust performance targets in data envelopment analysis with application to efficiency evaluation of airports[☆]

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ABSTRACT

We consider a ratio-based efficiency analysis in view of achieving performance targets by Decision Making Units. The targets refer to the efficiency scores or ranks and the truth of pairwise preference relations. These outcomes can be instantiated for at least one or all feasible weights associated with inputs and outputs. We discuss Multiple Objective Optimization problems that minimize the required reductions of consumed resources or increases of produced results needed to reach the pre-defined aims. The proposed models offer great flexibility to the Decision Maker, who can indicate which factors should be modified and to which extent. The evolutionary optimization algorithm computes a set of solutions that exhibit the trade-offs between expected modifications of various factors. They can serve as the basis for interactively selecting the most preferred solution to be implemented in practice. The use of the proposed framework is illustrated in the problem of analyzing the efficiency of Polish airports.

1. Introduction

Data Envelopment Analysis (DEA) is a non-parametric method focused on analyzing historical, objective data to control, monitor, and assess the performance of Decision Making Units (DMUs). The general aim of DEA consists of generating relative efficiency measures of DMUs in converting multiple inputs (resources) into multiple outputs (effects). This leads to the identification of the so-called efficient (best-practice) frontier. The units on the frontier are deemed efficient, whereas those below the frontier are classified as inefficient. DEA supports designing some corrective actions in the context of inefficient units (Cooper, Seiford, & Tone, 2007). DEA has been applied in thousands of real-world studies in areas including agriculture, banking, education, healthcare, transportation, and the public sector (Liu, Lu, Lu, & Lin, 2013).

Various models can be used to measure the relative efficiency in the presence of multiple inputs and outputs. Some commonly used include additive (Charnes, Cooper, Golany, Seiford, & Stutz, 1985), multiplicative (Charnes, Cooper, Seiford, & Stutz, 1982), and value-based (Almeida & Dias, 2012) models. However, the most popular

efficiency measure involves the ratio between virtual outputs and inputs, i.e., weighted sums of elementary outputs and inputs (Charnes, Cooper, & Rhodes, 1978). We consider such a model in this paper.

The primary reason underlying the widespread use and acceptance of ratio-based efficiency is its intuitiveness. Indeed, it extends the seminal definition of productivity by Farrell (1957) to handling multiple outputs and inputs, making it applicable to various contexts. Thus defined, efficiency increases with greater outputs and lesser inputs. Stakeholders can easily understand such dependencies without a deep understanding of mathematical or statistical concepts. Also, the respective weights indicate the contribution of each factor to the efficiency score. This allows Decision Makers (DMs) to understand which inputs or outputs are driving efficiency or its lack, aiding in identifying areas for improvement. By altering the weights, analysts can explore how resource allocation or results attainment changes impact efficiency. Overall, ratio-based scores provide a common metric for relative comparisons between DMUs, not requiring parametric modeling (i.e., assumptions about functional forms or production technologies).

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Moreover, they can be easily integrated into mathematical programming formulations used in DEA models, ensuring the feasibility of their application for real-world problems.

Even if DEA is greatly appreciated for its simplicity and generality, some criticism has also been formulated against how an efficiency analysis is performed in the traditional methods. Its main reason derives from a weak discrimination power (Podinovski & Thanassoulis, 2007), failing to handle uncertainty in data or its sensitivity to perturbation (Tavana, Toloo, Aghayi, & Arabmaldar, 2021), and neglecting the outcomes obtained for a large share of feasible weight vectors (Salo & Punkka, 2011; Zhang, Xia, Yang, Song, & Ang, 2021). Various research streams have been developed to prevent the drawbacks mentioned above. One of the most successful is related to Robustness Analysis (RA), which, in general, accounts for the uncertainties observed in the actual decision situations already at the stage of developing the model, deriving a recommendation for a particular problem, and verifying its stability (Kandakoglu, Walthers, & Amor, 2022). However, the uncertainty can be interpreted differently.

On the one hand, a significant body of research has been devoted to applying robust optimization (Ben-Tal, Ghaoui, & Nemirovski, 2009) in DEA. Its central focus is ensuring robustness in efficiency analysis in the presence of uncertain data (Toloo & Mensah, 2019). Since the inception of this hybrid approach in 2008 (Sadjadi & Omrani, 2008), various robust models have been proposed, dealing with different uncertainty sets and scenarios (for a review, see Peykani, Mohammadi, Saen, Sadjadi, and Rostamy-Malkhalifeh (2020) and Tavana et al. (2021)). The recent notable advances in this stream include simultaneous consideration of uncertainties in the inputs and outputs data and decreasing the computational burden of the original models (Toloo & Mensah, 2019) and proposing models for computing the extreme efficiency scores under various uncertainty sets (Salahi, Toloo, & Hesabirad, 2019). Other works addressed the problems of the potential lack of any efficient unit in the original robust optimization DEA (Toloo, Mensah, & Salahi, 2022), deriving robust cross-efficiency measures in the presence of uncertain data and undesirable outputs (Tavana et al., 2021), and identifying the underperforming DMUs according to the worst-practice frontier (Arabmaldar, Kwasi Mensah, & Toloo, 2021).

On the other hand, RA has been coupled with DEA to reflect the stability of efficiency results given the uncertainty related to the multiplicity of input and output weights. Thus understood RA is consistent with the definition formulated in the seminal paper by Roy (2010). According to his interpretation, RA needs to consider various versions of the problem formulation. Each version represents a reality that should be considered, reflecting a combination of the options related to the model's frailty points. In this regard, Salo and Punkka (2011) proposed the framework exploiting the variety of outcomes that can be obtained for the problem versions represented by all feasible weights. This method has been subsequently revised in Kadziński, Labijak, and Napieraj (2017). Overall, the available tools include the Linear Programming (LP) or Mixed-Integer Linear Programming (MILP) models that quantify the outcomes holding for at least one or all feasible weight vectors or the most and the least favorable settings for each DMU. These results include the minimal and maximal efficiency scores, the extreme efficiency ranks, and the necessary and possible efficiency preference relations. They can be considered robust with respect to all feasible weights. When the latter ones define the relevant problem versions, each unit always attains an efficiency score and a rank between the extreme (minimal and maximal) bounds, and the pairs related by the necessary relation are compared in the same way. The above approaches conducting RA assumed the performances of DMUs are certain. Recently, they have been adjusted to handle imprecise data (Labijak-Kowalska, Kadziński, Sychała et al., 2023), DMUs with fixed-sum outputs (Chen, Guo, Li, Liang, & Salo, 2021) or complex inner structure (Lei, Li, & Morton, 2022), an additive value efficiency model (Labijak-Kowalska & Kadziński, 2023), and a hierarchical structure of factors (Labijak-Kowalska, Kadziński and Mrozek,

2023). This paper extends the methods from the RA-oriented stream where uncertainty is related to selecting a particular weight vector for assessing the efficiency of DMUs.

Other criticism that can be leveled against the traditional DEA methods derives from providing straightforward solutions for improving efficiency. For example, they may take the form of a multiplier that must be applied simultaneously to all input or output values. Moreover, they focus only on becoming efficient, i.e., attaining a perfect efficiency score of one. In this regard, they lack flexibility in formulating performance targets. It is impossible to consider targets related to efficiencies lower than one, attaining arbitrary ranks, validating the truth of pairwise preference relations, or accounting for different levels of certainty. They also lack the means to exploit the variety of solutions that would lead to achieving the designated aims. In this perspective, it is impossible to account for the trade-offs between modification required on various inputs and/or outputs. Improving efficiency has also been neglected in the context of RA. The available tools focus only on quantifying the stability of results without involving formalized procedures that would delineate concrete scenarios leading a given DMU to better results.

1.1. The use of multiple objective optimization in data envelopment analysis

This paper aims to strengthen the links and inter-relations between DEA and Multiple Objective Optimization (MOO). This trend has been exploited in the existing literature in numerous ways.

In Färe, Grosskopf, and Lovell (1985), a non-radial measure of efficiency has been proposed. The underlying idea is that it is possible to find an individual multiplier for reducing the consumption of each resource or expanding the production of each outcome. In this regard, it does not involve equiproportional changes of inputs or outputs as in the radial DEA models. However, instead of individually optimizing these technical components, they are aggregated into a single non-radial measure called the Russell Graph Measure of Technical Efficiency. This idea has been subsequently extended in, e.g., (Pastor, Ruiz, & Sirvent, 1999), where one proposed the non-radial enhanced Russell model. It is defined as the ratio between an average decrease in input components and an average increase in output components.

Furthermore, Lins, Angulo-Meza, and Silva (2004) proposed two Multiple Objective Linear Programming (MOLP) models to determine non-radial projections for inefficient DMUs onto the Pareto efficient frontier. The first one considers each input to be reduced and each output to be increased as a separate objective function to formulate a Multiple Objective Ratio Optimization model. By solving it, they generate a set of optimal solutions, allowing the DM to choose the most preferred way to bring an inefficient DMU to the efficiency frontier. The other method, called Multiple Objective Target Optimization, incorporates preference information concerning the relative importance of optimized factors in the form of weights. They may be specified *a priori* or elicited in the interactive process using the interactive reference point technique called Pareto Race. Target input and output performances are presented to the DM so that they can test different weight sets to obtain the most preferred solution.

Another interactive approach proposed by Golany (1988) addresses the problem concerning output-oriented improvement where the output weights are unknown. The proposed interactive MOLP technique generates a set of alternative efficient solutions and then presents them to the DM, who is expected to pick the most preferred one. The process is repeated until a satisfactory solution is found. The new preference information provided by the DM in each iteration is used to adjust the direction of the objective function to produce more appropriate solutions.

In turn, Lozano and Villa (2009) presented two multiple objective DEA approaches for target setting, distinguished by the use of the AHP method (Saaty, 1990) to identify and differentiate the desired performance modifications. The first one is an interactive method, in which

Table 1
Types of robust results considered in the proposed framework.

Result	Type	Comparison	Character	Technique	Number of variables	
					Continuous	Binary
Efficiency score	Minimal	One vs. all	Cardinal	LP	$M + N$	–
	Maximal	One vs. all	Cardinal	MILP	$M + N$	K
Efficiency rank	Worst	One vs. all	Ordinal	MILP	$M + N$	$K - 1$
	Best	One vs. all	Ordinal	MILP	$M + N$	$K - 1$
Pairwise relation	Necessary	One vs. one	Ordinal	LP	$M + N$	–
	Possible	One vs. one	Ordinal	LP	$M + N$	–

the DM indicates the inputs and outputs to be improved, those for which current performances should be maintained, and those that can deteriorate. After obtaining the candidate target, the DM can provide new preference information to guide the optimization process or decide that the generated solutions meet their expectations. Conversely, the other method requires the DMs to specify their preferences, i.e., inputs and outputs priority levels, *a priori*. Then, it solves a series of DEA models (one for each priority level) to find a projection of an inefficient DMU onto the efficient frontier.

1.2. Aim of the paper

In this paper, we propose a method that considers the robust performance targets in the context of DEA. Its scheme is provided in Fig. 1.

The data required for its application involves the sets of DMUs, inputs, and outputs, the respective performances, and optional DM's preferences in the form of weight constraints. Such data is used to determine various types of robust efficiency results by solving dedicated mathematical programming models that exploit a set of all feasible weights. The analysis of such outcomes may stimulate the DMs to specify, for some DMU, a target that it cannot achieve with the current performances. Also, they should indicate which inputs and/or outputs can be changed and the possible constraints on the allowed performance modifications.

The considered targets may refer to the efficiency scores or ranks or the truth of pairwise preference relations in the context of all or at least one feasible weight vector. For example, one can account for the required modifications of selected inputs that need to be decreased minimally to attain any rank on the podium possibly. Alternatively, we may consider the minimal improvements of chosen outputs that would grant a given DMU the necessary or possible preference over some other DMU. Furthermore, one could study the reduction of some input and improvement of some output that would allow a DMU to attain an efficiency score of 0.8, 0.9, or 1 for all feasible weights. Thus, it is possible to consider targets differing in both the performance aim and certainty of their attainment.

Our main contribution consists of proposing MOO models for determining the minimal performance modifications required for attaining the designated performance targets. To solve the proposed models, we use an evolutionary MOO algorithm called NSGA-II (Deb, Pratap, Agarwal, & Meyarivan, 2002). Its application results in various solutions exhibiting different trade-offs between performance modifications on the considered inputs and/or outputs. In this way, the DM can exploit a variety of options for reaching a pre-defined target. Among such solutions, in the end, one should select the one that will be implemented in practice. Such a choice may be supported by some distance metric (e.g., L -norm) or an *a posteriori* MOO algorithm, which incorporates the DM's preferences. In this regard, we employ an interactive cone contraction method (Kadziński & Słowiński, 2012).

We illustrate the use of proposed models in a real-world problem concerning the evaluation of Polish airports. We consider three inputs related to the capacities of a terminal, runways, and an apron. In turn, the outputs concern passenger traffic and the number of aircraft movements. We identify the underperforming airports and investigate which

improvements would lead them to attain higher efficiency scores, more favorable ranks, or being preferred to another airport.

The paper is organized in the following way. Section 2 reminds the existing methods for robustness analysis using a ratio-based efficiency model. In Section 3, we introduce the MOO models for target achievements related to efficiency scores, ranks, and pairwise relations. Section 4 illustrates the use of the proposed approach to assess the efficiency of Polish airports. The last section summarizes the main contributions and delineates some directions for future research.

2. Reminder on robustness analysis methods in data envelopment analysis

In this section, we remind the methods for robustness analysis using a ratio-based efficiency model. These approaches quantify the stability of results given the multiplicity of feasible input and output weights in view of three perspectives related to the efficiency scores and ranks as well as pairwise preference relations. Let us use the following notation:

- $D = \{DMU_1, \dots, DMU_k, \dots, DMU_K\}$ – a set of K DMUs,
- x_m – m th input, $m \in \{1, \dots, M\}$, such that $x_{m,k}$ is a non-negative amount of m th input consumed by DMU_k ,
- y_n – n th output, $n \in \{1, \dots, N\}$, such that $y_{n,k}$ is a non-negative amount of n th output produced by DMU_k ,
- $v = (v_1, \dots, v_M)$ – a vector of non-negative input weights and $u = (u_1, \dots, u_N)$ – a vector of non-negative output weights.

The space of feasible input and output weights, denoted by S_v and S_u , respectively, is defined as follows:

$$S_v = \left\{ v = (v_1, \dots, v_M)^T \neq 0 \mid v \geq 0, A_v v \leq 0 \right\},$$

$$S_u = \left\{ u = (u_1, \dots, u_N)^T \neq 0 \mid u \geq 0, A_u u \leq 0 \right\}, \quad (1)$$

where A_v and A_u are coefficient matrices derived from the preference information elicited from the DM. The characteristics of the considered robust results and the underlying computational techniques are provided in Table 1.

2.1. Efficiency scores

To determine the efficiency of a DMU, Charnes et al. (1978) proposed the efficiency measure defined as a ratio of a weighted sum of outputs (virtual output) to a weighted sum of inputs (virtual input). Such an efficiency ratio is sought to be maximized and has an imposed limit of at most one. The above definition means that finding the (maximal) efficiency of a DMU requires solving the linear-fractional programming problem. By applying the Charnes–Cooper transformation, the respective model can be written as follows:

$$\begin{aligned} \text{Maximize} \quad & E_o^* = \sum_{n=1}^N u_n y_{no} \\ \text{subject to:} \quad & [CE_1^*] \sum_{m=1}^M v_m x_{mo} \leq 1, \\ & [CE_2^*] \sum_{n=1}^N u_n y_{nk} \leq \sum_{m=1}^M v_m x_{mk}, \quad k = 1, \dots, K, \\ & [CE_3^*] (v, u) \in (S_v, S_u). \end{aligned} \quad (2)$$

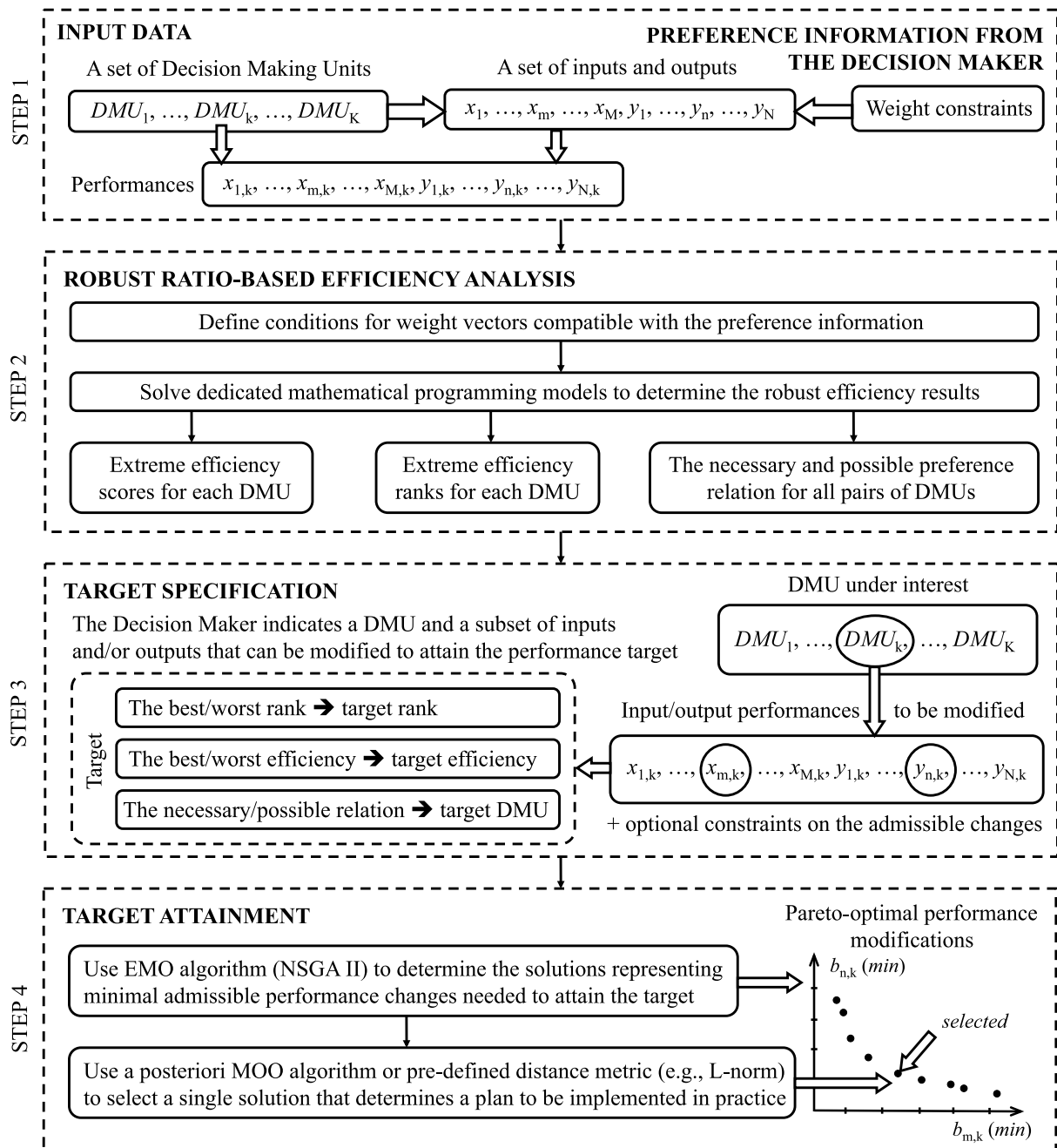


Fig. 1. The scheme of the proposed method.

For each $DMU_o \in D$, we search for the most favorable feasible weight vector $(v, u) \in (S_v, S_u)$ for which DMU_o achieves the highest possible efficiency, denoted by E_o^* . If E_o^* is equal to one, DMU_o is efficient; otherwise, it is inefficient.

Note that in the original linearized version of the problem, constraint $[CE_1^*]$ has the form of equality, i.e., we enforce the virtual input (the denominator in linear-fractional form) to be equal to one (Charnes et al., 1978). However, Toloo (2014) noted that this constraint can be relaxed to an inequality form and proved using the dual version of the problem model that the optimal solution remains identical. This modification of the original DEA CCR problem is particularly important

when dealing with uncertain input and output performances of DMUs. In such a case, equality constraints containing uncertain parameters may restrict the feasible region and even lead to infeasibility (Ben-Tal et al., 2009). Within the framework of our study, the performances of DMUs are assumed to be certain. Nevertheless, changing this constraint into inequality results in a less restrictive model and may reduce the computational burden. Therefore, we use a similar transformation when computing the minimal efficiency for each DMU.

To determine the efficiency score that the DMU achieves in the worst-case scenario (i.e., for the least favorable input and output weight vector), we consider the following MILP problem (Kadziński et al.,

2017):

$$\begin{aligned}
 &\text{Minimize} \quad E_{o,*} = \sum_{n=1}^N u_n y_{no} \\
 &\text{subject to:} \quad [CE_{*,1}] \sum_{m=1}^M v_m x_{mo} \geq 1, \\
 &\quad [CE_{*,2}] \sum_{n=1}^N u_n y_{nk} \geq \sum_{m=1}^M v_m x_{mk} - C(1 - z_k), \quad k = 1, \dots, K, \\
 &\quad [CE_{*,3}] \sum_{k=1}^K z_k \geq 1, \\
 &\quad [CE_{*,4}] z_k \in \{0, 1\}, \quad k = 1, \dots, K, \\
 &\quad [CE_{*,5}] (v, u) \in (S_v, S_u),
 \end{aligned} \tag{3}$$

where C denotes a large positive constant. When minimizing the efficiency score of DMU_o (denoted by $E_{o,*}$) in the set of feasible weights (constraint $[CE_{*,5}]$), we ensure that at least one $DMU_o \in D$ is efficient, i.e., its efficiency score is not less than one (see constraints $[CE_{*,2}] - [CE_{*,4}]$) (Kadziński et al., 2017).

2.2. Efficiency ranks

The rank of DMU_o relative to other DMUs over the vector (v, u) of input/output weights is defined as $R_o(v, u) = 1 + |R_k^>(v, u)|$, where $R_k^>(v, u) = \{k \in \{1, \dots, K\} \mid E_k(v, u) > E_o(v, u)\}$. In other words, $R_o(v, u)$ derives from the number of DMUs whose efficiency measures are strictly higher than that of DMU_o . The best rank $R_o^* = \min_{(v, u) \in (S_v, S_u)} R_o(v, u)$ that DMU_o can achieve is determined by solving the following MILP (Salo & Punkka, 2011):

$$\begin{aligned}
 &\text{Minimize} \quad R_o^* = 1 + \sum_{k=1, k \neq o}^K z_k \\
 &\text{subject to:} \quad [CR_1^*] \sum_{n=1}^N u_n y_{no} = \sum_{m=1}^M v_m x_{mo} = 1, \\
 &\quad [CR_2^*] \sum_{n=1}^N u_n y_{nk} \leq \sum_{m=1}^M v_m x_{mk} + C z_k, \quad k = 1, \dots, K, k \neq o, \\
 &\quad [CR_3^*] z_k \in \{0, 1\}, \quad k = 1, \dots, K, k \neq o, \\
 &\quad [CR_4^*] (v, u) \in (S_v, S_u).
 \end{aligned} \tag{4}$$

Constraint $[CR_1^*]$ enforces the virtual input and output, and thus the efficiency of DMU_o are all equal to one. To satisfy constraint $[CR_2^*]$, for each $DMU_k \in D \setminus DMU_o$, a corresponding binary variable z_k is instantiated with one if $E_k(v, u) > 1$ (i.e., the virtual output of DMU_k is greater than its virtual input). Under such conditions, DMU_k is ranked better than DMU_o . Otherwise, z_k is instantiated with zero. To determine the best rank attained by DMU_o , the solver minimizes the number of DMUs with more favorable efficiencies than DMU_o by assigning zeros to as many binary variables z_k as possible (Kadziński et al., 2017; Salo & Punkka, 2011).

Let us emphasize that we use equality constraints in all rank- or pairwise preference-oriented models. As opposed to cardinal efficiency scores, these results verify the ordinal properties (e.g., being more efficient or at least as good as other DMUs). Moreover, the formal proofs of these models' correctness are presented in Salo and Punkka (2011).

To determine the worst rank $R_{o,*} = \max_{(v, u) \in (S_v, S_u)} R_o(v, u)$ for DMU_o , the following MILP model needs to be solved (Salo & Punkka,

2011):

$$\begin{aligned}
 &\text{Maximize} \quad R_{o,*} = 1 + \sum_{k=1, k \neq o}^K z_k \\
 &\text{subject to:} \quad [CR_{*,1}] \sum_{n=1}^N u_n y_{no} \\
 &\quad = \sum_{m=1}^M v_m x_{mo} = 1, \\
 &\quad [CR_{*,2}] \sum_{m=1}^M v_m x_{mk} \\
 &\quad \leq \sum_{n=1}^N u_n y_{nk} + C(1 - z_k), \quad k = 1, \dots, K, k \neq o, \\
 &\quad [CR_{*,3}] z_k \in \{0, 1\}, \quad k = 1, \dots, K, k \neq o, \\
 &\quad [CR_{*,4}] (v, u) \in (S_v, S_u).
 \end{aligned} \tag{5}$$

It identifies a feasible vector $(v, u) \in (S_v, S_u)$ that maximizes the number of DMUs which are simultaneously not worse than DMU_o . Such units satisfy constraint $[CR_{*,2}]$ even when the expression $C(1 - z_k)$ is equal to zero (i.e., when their corresponding binary variables z_k are set to one). For DMUs with $E_k(v, u) < 1$ (i.e., efficiencies lesser than the efficiency of DMU_o), the binary variables are instantiated with zeros, thus preventing violation of $[CR_{*,2}]$. To determine the worst rank attained by DMU_o , the solver tries to assign ones to as many binary variables as possible (Kadziński et al., 2017; Salo & Punkka, 2011).

2.3. Pairwise efficiency preference relations

The analysis of the efficiency scores and ranks reveals how good a particular DMU is, relative to all DMUs in D . However, one may be interested in determining how efficient a DMU is when compared with another unit. When performing the pairwise comparisons of DMUs, it is meaningful to distinguish the following preference relations in the set D of DMUs (Kadziński et al., 2017):

- Possible efficiency preference relation $\succsim_E^P$, which holds for pairs of DMUs $(DMU_o, DMU_k) \in D \times D$ such that $E_o(v, u) \geq E_k(v, u)$ for at least one $(v, u) \in (S_v, S_u)$,
- Necessary efficiency preference relation $\succsim_E^N$, which holds for pairs of DMUs $(DMU_o, DMU_k) \in D \times D$ such that $E_o(v, u) \geq E_k(v, u)$ for all $(v, u) \in (S_v, S_u)$.

To verify the truth of these relations for a pair $(DMU_o, DMU_k) \in D \times D$, the following LP problem needs to be considered (Kadziński et al., 2017):

$$\begin{aligned}
 &\text{Maximize/Minimize} \quad E_o = \sum_{n=1}^N u_n y_{no} \\
 &\text{subject to:} \quad [CP_1] \sum_{m=1}^M v_m x_{mo} = 1, \\
 &\quad [CP_2] \sum_{n=1}^N u_n y_{nk} = \sum_{m=1}^M v_m x_{mk}, \\
 &\quad [CP_3] (v, u) \in (S_v, S_u).
 \end{aligned} \tag{6}$$

Constraint $[CP_2]$ imposes that the virtual input and output of DMU_k are equal, and its efficiency score is one. To verify whether DMU_o is possibly preferred to DMU_k , we maximize its virtual output while assuming that its virtual input is equal to one (constraint $[CP_1]$). If the obtained result is greater or equal to one, there exists at least one feasible vector $(v, u) \in (S_v, S_u)$ for which $E_o(v, u) \geq E_k(v, u)$, hence implying $DMU_o \succsim_E^P DMU_k$. The truth of the necessary efficiency preference relation is verified by minimizing the virtual output of DMU_o subject to the same constraint set. If the obtained result is not lesser than one, there is no feasible weight vector $(v, u) \in (S_v, S_u)$ for which $E_k(v, u) > E_o(v, u)$, hence implying $DMU_o \succsim_E^N DMU_k$ (Kadziński et al., 2017).

3. Multiple objective optimization models for the attainment of robust targets in data envelopment analysis

This section is dedicated to finding the optimal performance modifications leading to such an efficiency improvement that would allow achieving specific performance targets, as well as the maximal admissible deteriorations that would allow maintaining the target achieved with current performances. A similar approach, called Post Factum Analysis (PFA), has been proposed for Multiple Criteria Decision Aiding (MCDA) problems. Until now, PFA has been coupled with value- (Ciomek, Ferretti, & Kadziński, 2018), outranking- (Kadziński, Stamenković, & Uniejewski, 2022), and distance-based (Dutta, Singha, Goh, Lamata, & Verdegay, 2019) models. The framework proposed in this paper can be seen as an original adaptation of the PFA to the context of DEA.

In the remainder of this section, we provide a detailed description of the proposed approach. First, we identify the different types of targets. Next, we explain how the performances of DMUs can be modified. Then, we formalize the MOO problems for minimizing performance changes needed to achieve or maintain the target. Finally, we explain which algorithm we used to solve these problems.

3.1. Specification of performance targets

It is meaningful to consider the following types of performance targets:

- DMU_o achieves an efficiency score greater than or equal to the specified threshold value e possibly or necessarily, denoted by $E_o^* \geq e$ and $E_{o,*} \geq e$, respectively; in particular, one may be interested in determining the minimum improvements that are required for DMU_o to become efficient in a traditional DEA sense ($E_o^* \geq 1$);
- DMU_o attains an efficiency rank no worse than r for at least one or all feasible weight vectors, denoted by $R_o^* \leq r$ and $R_{o,*} \leq r$;
- DMU_o is possibly (necessarily) preferred to DMU_k , denoted by $DMU_o \gtrsim_E^P DMU_k$ ($DMU_o \gtrsim_E^N DMU_k$).

These targets concern three different perspectives of the efficiency analysis. Let us denote the general target as $T(X)$, where $X \in \{E_{o,*} \geq e, E_o^* \geq e, R_o^* \leq r, R_{o,*} \leq r, DMU_o \gtrsim_E^P DMU_k, DMU_o \gtrsim_E^N DMU_k\}$ represents the considered target result. We use $T(X) = 1$ to indicate the target's achievement. Otherwise, we write $T(X) = 0$.

3.2. Definition of performance changes

When the efficiency of the considered DMU is insufficient to achieve the designated target, it should improve current performances by reducing the excessive use of inputs and/or increasing the production of outputs. In turn, when the target is achieved, one may wonder how much can DMU deteriorate its current performances to maintain it. This section examines the constraints that must be satisfied by these modifications.

First, we need to define how the performances of the DMU are changed. Let $b_{x_{m,o}}$ denote the modification of performance of $DMU_o \in D$ on m th input and $b_{y_{n,o}}$ the change on n th output. Then $b(DMU_o) = [b_{x_{1,o}}, \dots, b_{x_{M,o}}, b_{y_{1,o}}, \dots, b_{y_{N,o}}]$ is the vector of performance modifications which, when applied to the performances of DMU_o , results in the construction of a new DMU_o^b . By applying changes, we mean reducing the consumption of each input $x_{m,o}$ by the appropriate value $b_{x_{m,o}}$, $m \in \{1, \dots, M\}$ and increasing the production of each output $y_{n,o}$ by $b_{y_{n,o}}$, $n \in \{1, \dots, N\}$. If the target is yet to be achieved, the performances of DMU_o need to be improved, so the performance modifications are positive. If our goal is to determine the maximum admissible deterioration for which the achieved target is maintained, the performance modifications are less than zero. Therefore, applying

them to DMU_o 's performances means an admissible increase in input consumption and a tolerated decrease in the production of outputs.

We shall consider two types of constraints to determine what performance modifications are feasible. On the one hand, the amount of each input (output) consumed (produced) by DMU_o must be within the acceptable range $[x_{m,o}^{min}, x_{m,o}^{max}]$ ($[y_{n,o}^{min}, y_{n,o}^{max}]$). For some problems, the lower and upper bounds of the performances are pre-defined. Otherwise, they are derived from the extreme performances observed in the dataset. This means that modification of performance of DMU_o on m th input $b_{x_{m,o}}$ must be within the range $[x_{m,o} - x_{m,o}^{max}, x_{m,o} - x_{m,o}^{min}]$, whereas for the n th output $b_{y_{n,o}} \in [y_{n,o}^{min} - y_{n,o}, y_{n,o}^{max} - y_{n,o}]$.

One can further restrict the allowed changes, e.g., by narrowing down the acceptable range of modifications or even completely excluding the possibility of changes on a subset of inputs and outputs. Furthermore, since the proposed framework allows adjusting the range of possible modifications, one may improve some of the DMU's performances while allowing others to deteriorate. This makes the proposed framework more general. In particular, it can be applied to more unusual cases, e.g., when the DMU is expected to produce more outputs, but it cannot be done without increasing the consumption of one of the inputs.

Taking into account the above restrictions, let us define the space of allowed performance modifications that can be applied to DMU_o as:

$$B(DMU_o) = \{b(DMU_o) \mid \forall i \in \{x_1, \dots, x_M\} \cup \{y_1, \dots, y_N\} \ b_{i,o}^{min} \leq b_{i,o} \leq b_{i,o}^{max}\}, \quad (7)$$

where $b_{i,o}^{min}$ and $b_{i,o}^{max}$ denote the minimum and the maximum changes, respectively, that can be applied to a particular input/output of DMU_o . Additionally, we denote the set of all performance modification vectors $b(DMU_o) \in B(DMU_o)$ for which $T(X) = 1$ by $B(T(X))$.

In the basic setting, we assume that the performances of the remaining units DMU_k , $k = 1, \dots, K$, $k \neq o$, remain unchanged. This is consistent with the hypothesis underlying the traditional DEA models, where the analysis of the actual performance of other units allows determining a hypothetical comparison unit or a benchmark to be followed by some inefficient DMU. In the eAppendix (supplementary material available online), we consider a setting where this assumption is relaxed.

3.3. Multiple objective optimization models for minimizing performance changes required for attaining the pre-defined targets

Having defined different types of performance targets and the way in which the performances of DMUs can be altered, let us analyze how to modify the performances optimally. In the case of simultaneous optimization of several factors, typically, a change in one can be compensated by an appropriate modification in another. This means that, typically, it is possible to achieve the target in many different ways. Clearly, when the target cannot be achieved with current performances, we expect the required improvements to be as small as possible. In turn, when the target is already achieved, we aim to maximize the deterioration for which the target is still maintained. Nevertheless, the definition of performance modifications provided in the previous section allows both cases to be considered as the problem of minimizing the improvements required to achieve the target. If the target is to be maintained, the minimal improvements that are smaller than zero are equivalent to the maximal admissible deteriorations. In this section, we propose MOO models for determining solutions that minimize the performance modifications required to achieve or maintain the targets related to efficiency scores, ranks, and preference relations.

3.3.1. Attaining targets related to efficiency scores

When considering an inefficient DMU, one may ask how to improve its performances to make it efficient. In turn, once the DMU proved efficient, one may wonder what maximal performance deterioration would allow this DMU to remain efficient. More general questions can be formulated as follows: what minimal performance modifications would allow DMU_o to attain the efficiency score that is not less than a certain threshold e or, in other words, how to achieve target $T(E_o^* \geq e)$? To solve this problem, we propose the following MOO model:

$$\begin{aligned}
 &\text{Minimize} && b_{x_{1,o}}, \dots, b_{x_{M,o}}, b_{y_{1,o}}, \dots, b_{y_{N,o}} && [TE^*] \\
 &\text{subject to:} && [CTE_{*,1}^*] \quad b_{i,o}^{\min} \leq b_{i,o} \leq b_{i,o}^{\max}, && i \in \{x_1, \dots, x_M\} \cup \{y_1, \dots, y_N\} \\
 &&& [CTE_{*,2}^*] \quad \exists(v, u) \in (S_v, S_u) \\
 &&& \sum_{n=1}^N u_n(y_{n,o} + b_{y_{n,o}}) \geq e, \\
 &&& [CTE_{*,3}^*] \quad \sum_{m=1}^M v_m(x_{m,o} - b_{x_{m,o}}) \leq 1, \\
 &&& [CTE_{*,4}^*] \quad \sum_{n=1}^N u_n(y_{n,o} + b_{y_{n,o}}) \\
 &&& \leq \sum_{m=1}^M v_m(x_{m,o} - b_{x_{m,o}}), \\
 &&& [CTE_{*,5}^*] \quad \sum_{n=1}^N u_n y_{n,k} \\
 &&& \leq \sum_{m=1}^M v_m x_{m,k}, && k = 1, \dots, K, \quad k \neq o, \\
 &&& [CTE_{*,6}^*] \quad (v, u) \in (S_v, S_u).
 \end{aligned}$$

The solution to the above problem, denoted by $B^{PF}(T(E_o^* \geq e))$, is the set of Pareto optimal performance modification vectors. Each vector $b(DMU_o) \in B^{PF}(T(E_o^* \geq e))$ belongs to the space of admissible performance changes $B(DMU_o)$, which is guaranteed by constraint $[CTE_{*,1}^*]$. Constraints $[CTE_{*,2}^*] - [CTE_{*,6}^*]$ ensure that for each vector $b(DMU_o) \in B^{PF}(T(E_o^* \geq e))$, target $T(E_o^* \geq e)$ is satisfied, i.e., there exists at least one feasible weight vector $(v, u) \in (S_v, S_u)$, for which the efficiency of DMU_o is greater than or equal to e . In traditional DEA, DMUs that attain the maximum efficiency score of one are considered efficient. However, their efficiencies measured under the most unfavorable conditions may be significantly lower. To obtain a universal DMU performing well under any conditions, we should consider the target related to the minimum efficiency score. Although the target requiring DMU_o to be efficient for all feasible weight vectors may be challenging to achieve, one may choose another reasonable value $e < 1$ and designate target $T(E_o^* \geq e)$. To determine what minimal improvements would allow achieving or maintaining this target, we consider the following MOO problem:

$$\begin{aligned}
 &\text{Minimize} && b_{x_{1,o}}, \dots, b_{x_{M,o}}, b_{y_{1,o}}, \dots, b_{y_{N,o}} && [TE_*] \\
 &\text{subject to:} && [CTE_{*,1}] \quad b_{i,o}^{\min} \leq b_{i,o} \leq b_{i,o}^{\max}, && i \in \{x_1, \dots, x_M\} \cup \{y_1, \dots, y_N\} \\
 &&& [CTE_{*,2}] \quad \exists(v, u) \in (S_v, S_u) \\
 &&& \sum_{n=1}^N u_n(y_{n,o} + b_{y_{n,o}}) < e, \\
 &&& [CTE_{*,3}] \quad \sum_{m=1}^M v_m(x_{m,o} - b_{x_{m,o}}) \geq 1, \\
 &&& [CTE_{*,4}] \quad \sum_{n=1}^N u_n(y_{n,o} + b_{y_{n,o}}) \\
 &&& \geq \sum_{m=1}^M v_m(x_{m,o} - b_{x_{m,o}}) - C(1 - z_o), \\
 &&& [CTE_{*,5}] \quad \sum_{n=1}^N u_n y_{n,k} \\
 &&& \geq \sum_{m=1}^M v_m x_{m,k} - C(1 - z_k), && k = 1, \dots, K, \quad k \neq o, \\
 &&& [CTE_{*,6}] \quad \sum_{k=1}^K z_k \geq 1, \\
 &&& [CTE_{*,7}] \quad z_k \in \{0, 1\}, && k = 1, \dots, K, \\
 &&& [CTE_{*,8}] \quad (v, u) \in (S_v, S_u).
 \end{aligned}$$

The solution to the above problem is a set of non-dominated performance modification vectors $B^{PF}(T(E_o^* \geq e)) \in B(T(E_o^* \geq e))$. For each $b(DMU_o) \in B^{PF}(T(E_o^* \geq e))$, there is no other vector in $B(T(E_o^* \geq e))$ that would allow achieving the target with lesser performance modifications. Constraints $[CTE_{*,2}]$ and $[CTE_{*,3}]$ ensure that there is no feasible weight vector $(v, u) \in (S_v, S_u)$ for which the efficiency score of DMU_o would be lower than e , which guarantees the achievement of target $T(E_o^* \geq e)$. Vector (v, u) is feasible if it belongs to (S_v, S_u) and there is at least one DMU (DMU_o or some

$DMU_k \in D \setminus \{DMU_o\}$) that attains an efficiency score equal to one (see constraints $[CTE_{*,4}] - [CTE_{*,8}]$).

3.3.2. Attaining targets related to efficiency ranks

In this section, we focus on the targets related to efficiency ranks. Let us consider DMU_o whose rank varies in the interval $[R_o^*, R_{o,*}]$. In this case, one may wonder what performance modifications would allow DMU_o to achieve a particular rank under the most favorable conditions or what changes are required for DMU_o to be ranked no worse than r th in the worst-case scenario. Alternatively, these questions may be formulated as problems of achieving or maintaining rank r possibly or necessarily.

To identify the minimal performance modifications leading DMU_o to be ranked r th for at least one feasible weight vector, i.e., to achieve or preserve the target $T(R_o^* \leq r)$, we need to solve the following MOO problem:

$$\begin{aligned}
 &\text{Minimize} && b_{x_{1,o}}, \dots, b_{x_{M,o}}, b_{y_{1,o}}, \dots, b_{y_{N,o}} && [TR^*] \\
 &\text{subject to:} && [CTR_{*,1}^*] \quad b_{i,o}^{\min} \leq b_{i,o} \leq b_{i,o}^{\max}, && i \in \{x_1, \dots, x_M\} \cup \{y_1, \dots, y_N\} \\
 &&& [CTR_{*,2}^*] \quad \exists(v, u) \in (S_v, S_u) \\
 &&& 1 + \sum_{k=1, k \neq o}^K z_k \leq r, \\
 &&& [CTR_{*,3}^*] \quad \sum_{n=1}^N u_n(y_{n,o} + b_{y_{n,o}}) \\
 &&& = \sum_{m=1}^M v_m(x_{m,o} - b_{x_{m,o}}) = 1, \\
 &&& [CTR_{*,4}^*] \quad \sum_{n=1}^N u_n y_{n,k} \\
 &&& \leq \sum_{m=1}^M v_m x_{m,k} + C z_k, && k = 1, \dots, K, \quad k \neq o, \\
 &&& [CTR_{*,5}^*] \quad z_k \in \{0, 1\}, && k = 1, \dots, K, \quad k \neq o, \\
 &&& [CTR_{*,6}^*] \quad (v, u) \in (S_v, S_u).
 \end{aligned}$$

The resulting set of performance modification vectors, denoted by $B^{PF}(T(R_o^* \leq r))$, provides optimal ways to attain the efficiency rank no worse than r for at least one feasible weight vector. Constraint $[CTR_{*,3}^*]$ imposes the efficiency ratio of DMU_o is equal to one. Further, for each $DMU_k \in D \setminus \{DMU_o\}$ that is more efficient than DMU_o , the corresponding binary variable z_k is instantiated with one, to prevent violation of constraint $[CTR_{*,4}^*]$. In this way, the sum of binary variables z_k indicates the number of DMUs that are more efficient than DMU_o . When increased by one, it is equal to the rank attained by DMU_o for a particular weight vector $(v, u) \in (S_v, S_u)$. We require that for at least one feasible weight vector, this rank is no worse than r (constraint $[CTR_{*,5}^*]$).

To find the minimal performance modifications that guarantee the necessary achievement of the r th rank by DMU_o , the following MOO model needs to be considered:

$$\begin{aligned}
 &\text{Minimize} && b_{x_{1,o}}, \dots, b_{x_{M,o}}, b_{y_{1,o}}, \dots, b_{y_{N,o}} && [TR_*] \\
 &\text{subject to:} && [CTR_{*,1}] \quad b_{i,o}^{\min} \leq b_{i,o} \leq b_{i,o}^{\max}, && i \in \{x_1, \dots, x_M\} \cup \{y_1, \dots, y_N\} \\
 &&& [CTR_{*,2}] \quad \exists(v, u) \in (S_v, S_u) \\
 &&& 1 + \sum_{k=1, k \neq o}^K z_k > r, \\
 &&& [CTR_{*,3}] \quad \sum_{n=1}^N u_n(y_{n,o} + b_{y_{n,o}}) \\
 &&& = \sum_{m=1}^M v_m(x_{m,o} - b_{x_{m,o}}) = 1, \\
 &&& [CTR_{*,4}] \quad \sum_{m=1}^M v_m x_{m,k} \\
 &&& \leq \sum_{n=1}^N u_n y_{n,k} + C(1 - z_k), && k = 1, \dots, K, \quad k \neq o, \\
 &&& [CTR_{*,5}] \quad z_k \in \{0, 1\}, && k = 1, \dots, K, \quad k \neq o, \\
 &&& [CTR_{*,6}] \quad (v, u) \in (S_v, S_u).
 \end{aligned}$$

By solving it, we obtain a set of non-dominated performance modification vectors $b(DMU_o) \in B^{PF}(T(R_o^* \leq r))$ thanks to which DMU_o would not be ranked worse than r th for any feasible weight vector. This is imposed by constraints $[CTR_{*,2}] - [CTR_{*,6}]$ ensuring there is no feasible weight vector $(v, u) \in (S_v, S_u)$ for which DMU_o would attain rank greater than r . Note that the rank is modeled as the sum of binary variables z_k , $k = 1, \dots, K$, $k \neq o$, increased by 1.

3.3.3. Attaining targets related to pairwise efficiency preference relations

In this section, we investigate the improvements that need to be made to attain preference-related targets. Let us start with the scenario where the robustness analysis revealed that some DMU_o is less efficient than DMU_k for all feasible weight vectors $(v, u) \in (S_v, S_u)$. In this case, one may wonder how much DMU_o should improve to become preferred over DMU_k for at least one weight vector. To achieve target $T(DMU_o \succ_E^P DMU_k)$ with minimal effort, we need to solve the following MOO problem:

$$\begin{aligned} &\text{Minimize} && b_{x_{1,o}}, \dots, b_{x_{M,o}}, b_{y_{1,o}}, \dots, b_{y_{N,o}} && [TP] \\ &\text{subject to:} && [CTP_1] \ b_{i,o}^{min} \leq b_{i,o} \leq b_{i,o}^{max}, && i \in \{x_1, \dots, x_M\} \cup \{y_1, \dots, y_N\} \\ &&& [CTP_2] \ \exists (v, u) \in (S_v, S_u) \\ &&& \sum_{n=1}^N u_n (y_{n,o} + b_{y_{n,o}}) \geq 1, \\ &&& [CTP_3] \ \sum_{m=1}^M v_m (x_{m,o} - b_{x_{m,o}}) \\ &&& = 1, \\ &&& [CTP_4] \ \sum_{n=1}^N u_n y_{n,k} \\ &&& = \sum_{m=1}^M v_m x_{m,k}, \\ &&& [CTP_5] \ (v, u) \in (S_v, S_u). \end{aligned}$$

Its solution amounts to a set of Pareto optimal vectors, denoted by $B^{PF}(T(DMU_o \succ_E^P DMU_k))$, modifying the performances of DMU_o so that it is preferred to DMU_k for at least one feasible weight vector. The achievement of this target is guaranteed by constraints $[CTP_2] - [CTP_5]$. They ensure that there exists some $(v, u) \in (S_v, S_u)$ such that $E_{DMU_o} \geq 1$, while $E_k = 1$, and therefore $DMU_o \succ_E^P DMU_k$.

Let us now consider a more demanding target $T(DMU_o \succ_E^N DMU_k)$ that requires DMU_o to be at least as efficient as DMU_k for all feasible weight vectors. The minimal changes sufficient to attain it can be determined with the following MOO problem:

$$\begin{aligned} &\text{Minimize} && b_{x_{1,o}}, \dots, b_{x_{M,o}}, b_{y_{1,o}}, \dots, b_{y_{N,o}} && [TN] \\ &\text{subject to:} && [CTN_1] \ b_{i,o}^{min} \leq b_{i,o} \leq b_{i,o}^{max}, && i \in \{x_1, \dots, x_M\} \cup \{y_1, \dots, y_N\} \\ &&& [CTN_2] \ \nexists (v, u) \in (S_v, S_u) \\ &&& \sum_{n=1}^N u_n (y_{n,o} + b_{y_{n,o}}) < 1, \\ &&& [CTN_3] \ \sum_{m=1}^M v_m (x_{m,o} - b_{x_{m,o}}) \\ &&& = 1, \\ &&& [CTN_4] \ \sum_{n=1}^N u_n y_{n,k} \\ &&& = \sum_{m=1}^M v_m x_{m,k}, \\ &&& [CTN_5] \ (v, u) \in (S_v, S_u). \end{aligned}$$

In the above problem, each performance modification vector $b(DMU_o) \in B^{PF}(T(DMU_o \succ_E^N DMU_k))$ ensures that DMU_o is necessarily preferred to DMU_k . This is guaranteed by constraints $[CTN_2] - [CTN_5]$ imposing there is no admissible weight vector $(v, u) \in (S_v, S_u)$ for which the efficiency of DMU_o is lower than the efficiency of DMU_k , and hence $DMU_o \succ_E^N DMU_k$.

3.4. An evolutionary approach for computing performance modifications needed for attaining the targets

In this section, we provide a detailed explanation of the method used to achieve the targets through minimal improvements. Each target can be attained by solving the dedicated optimization problem. However, since all multiple objective problems proposed in Section 3.3 are non-linear, we decided to use the popular evolutionary algorithm NSGA-II (Deb et al., 2002) and its open-source implementation available as R package (Mersmann, Trautmann, Steuer, Bischl, & Deb, 2020).

NSGA-II is a meta-heuristic algorithm that can solve complex optimization problems involving multiple objective functions and constraints. In our case, it is used to find a set of Pareto-optimal performance modification vectors that allow DMU_o to achieve a specific

target. NSGA-II starts with a parent population of randomly initialized solutions of size N . The solutions are the real-coded vectors of performance modifications belonging to the space of allowed performance modifications. Then, until the stopping condition is met, the algorithm performs the following steps. First, it creates offspring of the same size as the parent population using binary tournament selection, simulated binary crossover, and polynomial mutation (Deb et al., 2002). Then, it combines the parents and the offspring into one population. Finally, it creates a new parent population by choosing N best individuals based on two measures: non-dominance rank and crowding distance (Deb et al., 2002). The former criterion promotes solutions closer to the actual Pareto front, while the latter maintains diversity in the population.

Let us explain an approach we used to ensure that the constraints related to the target achievement are satisfied. First, let us recall the constraint-handling strategy proposed for NSGA-II. The primary criterion used by the algorithm to determine which parents are used for mating during the tournament selection and to select the best individuals is based on the dominance principle. In the case of a constrained MOO problem, each solution can be either feasible or infeasible. Then, according to the dominance principle, any feasible solution dominates any infeasible solution. Furthermore, when comparing a pair of feasible solutions, the one with a better objective function value is more advantageous. Finally, when both solutions are infeasible, we prefer the one with a smaller constraint violation (Deb et al., 2002).

In our case, the solution is feasible if the performance changes it represents are sufficient to achieve the designated target. As the targets concern the possible or necessary achievement of a specific efficiency score, rank, or preference relation, we need to determine whether the solution satisfies the appropriate constraints for at least one or all feasible weight vectors. For this purpose, we simulate the changes by applying a performance modification vector to DMU_o . Then, depending on the target type, we calculate the minimum (maximum) efficiency score or rank attained by DMU_o^b , or verify the truth of the appropriate efficiency preference relation. For this, we adapted the implementation of the models presented in Section 2 available within the open-source framework for robustness analysis (Kadziński et al., 2017). When compared with the target, the obtained result indicates whether the solution is feasible. Although this binary information is sufficient to determine the dominance relation for a pair of solutions where at least one of them is feasible, it is not well suited to compare pairs of infeasible solutions. In turn, we calculate the difference between the actual and expected efficiency scores or ranks. For the preference-related targets, we find the efficiency gap that needs to be covered to turn their falsity into the truth. As a result, the solutions close to achieving the target will dominate those violating the constraints significantly. This drives a population of solutions closer to the actual Pareto front.

4. Illustrative case study: analysis of efficiency of polish airports

This section illustrates the use of the proposed framework on real-world data concerning Polish airports (Kadziński et al., 2017). Measuring airport efficiency is vital from the viewpoint of airlines, governments, managers, and passengers (Yang, 2010). DEA methods are widely used for this purpose (De Oña, Cui, & Yu, 2021).

As summarized in Kadziński et al. (2017), the primary input types considered in this kind of application are related to terminal services (e.g., number of check-in desks or gates, and terminal or baggage claim area), movement model (e.g., the airport area, apron area, aircraft parking positions, numbers of runways, runway length), finance (e.g., operational costs, capital invested, and airport charge), labor (e.g., number of employees), and the airport's localization (e.g., distance to the nearest city center or catchment area). In turn, the major types of considered outputs refer to the terminal services (e.g., number of passengers or cargo throughput), movement model (e.g., aircraft or commuter movement, and number of air carrier operations), finance

Table 2
Input and output performances of Polish airports.

Airport	Short name	i_1	i_2	i_3	o_1	o_2
Warsaw	WAW	10.5	36	129.4	9.5	129.7
Cracow	KRK	3.1	19	31.6	2.9	31.3
Katowice	KAT	3.6	32	57.6	2.4	21.1
Wroclaw	WRO	1.5	12	18.0	1.5	18.8
Poznan	POZ	1.5	10	24.0	1.3	16.2
Lodz	LCJ	0.6	12	24.0	0.3	4.2
Gdansk	GDN	1.0	15	42.9	2	23.6
Szczecin	SZZ	0.7	10	25.7	0.3	4.2
Bydgoszcz	BZG	0.3	6	3.4	0.3	6.2
Rzeszow	RZE	0.6	6	11.3	0.3	3.5
Zielona Gora	IEG	0.1	10	63.4	0.005	0.61

(e.g., operational revenue, sales to planes, and non-aeronautical fee). This paper focuses on assessing technical efficiency, which measures how well an airport utilizes its resources to produce services in the context of passenger traffic. Therefore, we refer to the factors reflecting terminal services and a movement model. Specifically, we take into account the following three inputs (i_1 – i_3) and two outputs (o_1 – o_2):

- i_1 : an annual capacity of a terminal, defined as the flow of passengers that the airport can accommodate without causing serious inconvenience (units – millions of passengers per year),
- i_2 : a maximal runway capacity – an average number of landings and take-offs (movements) that can be carried out on the airport's runways (number of movements per hour),
- i_3 : a dynamic apron capacity – an average number of aircraft that can be accommodated by the airport (number of aircraft per hour),
- o_1 : passengers traffic – the total number of passengers served by the airport (millions of passengers per year),
- o_2 : the number of aircraft movements – the total number of landings and take-offs performed at the airport (thousands of movements per year).

The performances of all airports are provided in Table 2.

First, we report the robust results (i.e., efficiency scores, ranks, and pairwise preference relations). Then, they will serve as the starting point for analyzing the impact of performance changes on the efficiency outcomes.

4.1. Robustness analysis

4.1.1. Efficiency scores

In this section, we focus on the best and the worst efficiency scores obtained by each $DMU_o \in D$. These results are summarized in Table 3. Five airports (WAW, KRK, WRO, GDN, BZG) are considered efficient in the traditional DEA sense. Each attains the maximum efficiency score E_o^* equal to one, forming the so-called efficient frontier. The efficiency of the remaining airports is measured relative to the efficiency frontier. The greater the distance of a given airport from the efficient frontier, the lower its efficiency score. POZ has the smallest distance from the efficiency frontier among the six inefficient airports. Its efficiency score is equal to 0.8. The next is KAT, with an efficiency score of 0.59. The airport with the greatest gap to the efficiency frontier is IEG, with a maximum efficiency score of 0.258. In subsequent sections of this paper, we will examine what performance changes would be needed for these three inefficient airports to become efficient.

The minimum efficiency $E_{o,*}$ for all airports does not exceed 0.5. In particular, BZG is very sensitive to the choice of the weight vector, as its efficiency score in the worst-case scenario (0.189) is more than five times lower than its maximum efficiency score. Note that $E_{BZG,*}$ is close to the minimum efficiency attained by a highly inefficient airport KAT ($E_{KAT}^* = 0.59$). Furthermore, the minimum efficiency score of POZ is equal to 0.37, nearly twice as high as the worst score for the efficient BZG.

Table 3
Minimum and maximum efficiency scores and ranks obtained by airports.

Airport	E_o^*	$E_{o,*}$	R_o^*	$R_{o,*}$
WAW	1.000	0.452	1	5
KRK	1.000	0.428	1	6
KAT	0.590	0.183	6	10
WRO	1.000	0.435	1	5
POZ	0.800	0.370	3	6
LCJ	0.300	0.095	7	10
GDN	1.000	0.302	1	6
SZZ	0.271	0.090	8	10
BZG	1.000	0.189	1	8
RZE	0.409	0.162	7	11
IEG	0.258	0.001	8	11

Table 4
The necessary and possible efficiency preference relations.

Airport	Necessary preference	Possible preference
WAW	KAT, POZ, LCJ, SZZ, RZE, IEG	KRK, WRO, GDN, BZG
KRK	KAT, LCJ, SZZ, RZE, IEG	WAW, WRO, POZ, GDN, BZG
KAT	RZE	LCJ, SZZ, BZG, IEG
WRO	KAT, LCJ, SZZ, RZE, IEG	WAW, KRK, POZ, GDN, BZG
POZ	KAT, LCJ, SZZ, RZE, IEG	KRK, WRO, GDN, BZG
LCJ	IEG	KAT, SZZ, RZE
GDN	KAT, LCJ, SZZ, RZE, IEG	WAW, KRK, WRO, POZ, BZG
SZZ	–	KAT, LCJ, RZE, IEG
BZG	LCJ, SZZ, RZE, IEG	WAW, KRK, KAT, WRO, POZ, GDN
RZE	–	LCJ, SZZ, BZG, IEG
IEG	–	KAT, SZZ, RZE

4.1.2. Efficiency ranks

This section discusses the best and the worst efficiency ranks obtained by each $DMU_o \in D$ (see Table 3). They are derived from comparing DMUs' efficiencies pairwise. As a result, they have a straightforward interpretation — the best (worst) rank of DMU_o reduced by one indicates how many other DMUs are strictly better in terms of the efficiency score for the most (least) advantageous weight vector for DMU_o . In this perspective, the best possible efficiency rank R_o^* for the five airports previously deemed efficient needs to be equal to one. However, the ranks they attain in the worst-case scenarios are significantly different. While WAW and WRO are ranked no worse than fifth, BZG may drop to eighth place. Meanwhile, the rank of the inefficient airport POZ ranges from three to six, indicating that there are at least two and at most five other airports simultaneously (i.e., for the same feasible weight vector) more efficient than POZ. The best rank for KAT is six, and it is ranked in the penultimate place for the least favorable weight vector.

4.1.3. Pairwise efficiency preference relations

The efficiency scores and ranks give an insight into how efficient the individual DMUs are relative to the others. However, their intervals may sometimes be too broad to offer sufficiently decisive results. Moreover, they fail to indicate which of the two selected DMUs is more efficient, especially when their rank intervals overlap or both achieve an efficiency score of one in the most advantageous case. For this purpose, an additional analysis involving the comparison of each pair of DMUs is in order. The necessary and possible efficiency preference relations are presented in Table 4. The necessary relation is also illustrated by the Hasse diagram (see Fig. 2). Please note that by definition, the truth of necessary preference relation implies the truth of possible relation and that these two relations are reflexive.

Let us first consider the five airports deemed as efficient. Indeed, no airport in the dataset is necessarily preferred over any of them. However, only WAW is necessarily preferred over all inefficient airports. The remaining efficient airports (KRK, WRO, GDN, and BZG) are better than POZ for some weight vectors while being worse for others (i.e., POZ is possibly strictly preferred over them). In addition,

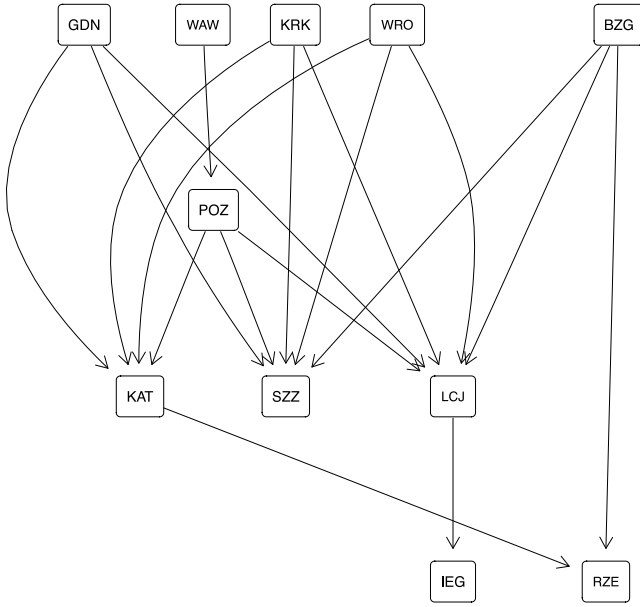


Fig. 2. The Hasse diagram representing the necessary efficiency preference relation.

POZ is necessarily preferred over the five remaining inefficient airports, whereas BZG is preferred over the inefficient KAT only in a possible way. Furthermore, even though BZG is necessarily preferred to RZE, there is at least one weight vector for which RZE is preferred to BZG. For such a vector, the efficiencies of both airports must be equal.

Let us examine the interdependencies between the necessary preference relations and efficiency ranks. On the one hand, the number of DMUs that are possibly preferred over DMU_o increased by one is the upper bound of the efficiency rank it can attain. For example, there are four airports that are possibly preferred over WAW, and its worst rank is fifth ($4 + 1 = 5$). In contrast, the worst rank of POZ is six because five other airports are possibly preferred over it. On the other hand, the best rank of DMU_o is bounded by the number of units that are necessarily preferred to it, and at the same time, DMU_o is not possibly preferred to them. For example, there are five airports that are necessarily preferred over KAT, and its best efficiency rank is six. However, POZ is, in the best case, ranked third, even though there is only one airport (WAW) that is necessarily preferred over it.

4.2. Performance improvements

Let us illustrate the method proposed in this paper by examining how some inefficient airports' performances need to be improved to achieve the designated targets. Although, in general, the proposed approach allows for the optimization of any number of factors, we analyze subsets of no more than three factors due to cognitive limitations.

In the main paper, we discuss the results for the targets related to the efficiency scores. The targets concerning ranks and preference relations are presented in the eAppendix. The optimization was performed using NSGA-II, which ran for 500 generations. The population consisted of 200 and 600 individuals for problems involving modifications on two and three factors, respectively. The space of admissible performance changes for each considered airport was delineated so that the modified performances would not exceed the extreme performances observed in the dataset.

In what follows, we analyze the performance modifications required to improve the efficiency scores of three airports judged inefficient (KAT, POZ, and IEG). We consider both the possible and necessary achievement of some expected efficiency scores. In addition, we show what improvements are needed for a particular airport to achieve

a series of increasingly demanding targets. For illustrative purposes, we modify various subsets of factors, including inputs, outputs, or a mix of both. For some scenarios, we show how the most preferred solution can be selected using either some default distance metric or an interactive MOO method. Also, for comparative purposes, we refer to the solutions obtained from the standard Charnes–Cooper–Rhodes (CCR) model (Charnes et al., 1978). This model is based on a radial projection of all inputs or all outputs of an inefficient unit in the context of attaining efficiency, i.e., the best score E^* of one for at least one feasible weight vector. For this reason, such a comparison is provided only for the scenarios involving the possible attainment of $E^* = 1$. The score-based targets considered in this section are listed in Table 5.

4.2.1. The necessary and possible achievement of targets

In this section, we demonstrate the difference between the possible and necessary achievement of the same efficiency score. Let us begin with KAT, whose efficiency score ranges from 0.183 to 0.590. The analysis revealed that this airport could become efficient (i.e., attain a maximum efficiency score of one) by reducing the amounts of used inputs while maintaining the current production of outputs. However, in the worst-case scenario, the highest efficiency score, which can be achieved by the maximum reduction of all inputs, is slightly lower ($E_{KAT,*} = 0.976$). Nevertheless, it would be interesting to compare the changes needed to achieve the same efficiency score possibly and necessarily. For this purpose, we specified the appropriate targets: $T(E_{KAT,*}^* \geq 0.95)$ and $T(E_{KAT,*} \geq 0.95)$.

The results presented in Fig. 3 indicate that there are many ways to improve the maximum efficiency score and achieve the first target. With so many optimal ways to improve, it is worth considering which could be preferred by the DM. Without any additional preference information, we can opt for solutions that optimize the distance between the original and modified performances of the airport. As the performance modification vector already represents the performance differences, we can determine that distance by calculating the norm of this vector. However, since the performances on different inputs and outputs were measured using different scales and units, we need to normalize these vectors first. For this purpose, we rescaled the performance modifications on each input and output to fall within a common range $[0, 1]$ by performing the following linear transformation:

$$b'_{i,o} = \frac{b_{i,o} - b_{i,normal}^{min}}{b_i^{max} - b_i^{min}}, \quad i \in \{x_1, \dots, x_M\} \cup \{y_1, \dots, y_N\},$$

For example, to achieve the target $T(E_{KAT,*}^* \geq 0.95)$, we can choose the vector that optimizes the L_1 norm, i.e., $[b_{i_1}, b_{i_2}, b_{i_3}] = [0.0, 0.07, 31.38]$. These changes involve maintaining the current capacity of a terminal (i_1) and a slight improvement in maximal runway capacity (i_2). In contrast, the dynamic apron capacity (i_3) needs to be reduced significantly. Alternatively, we may use the L_2 norm, which penalizes significant modifications, thus promoting more balanced vectors. The best solution in terms of this norm $[b_{i_1}, b_{i_2}, b_{i_3}] = [1.70, 0.03, 16.07]$ requires less improvement on i_3 at the expense of reducing the terminal capacity (i_1). Finally, if our goal is to minimize the maximal required (normalized) performance improvement on each input, we should choose the vector $[b_{i_1}, b_{i_2}, b_{i_3}] = [1.46, 13.65, 19.95]$ that optimizes the L_∞ norm.

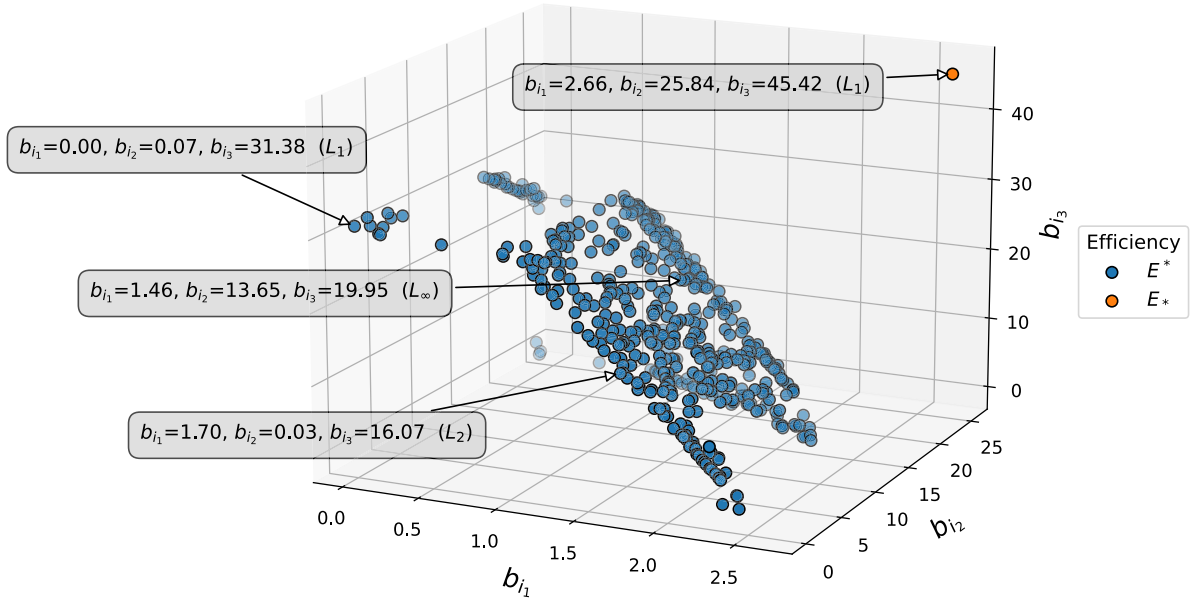
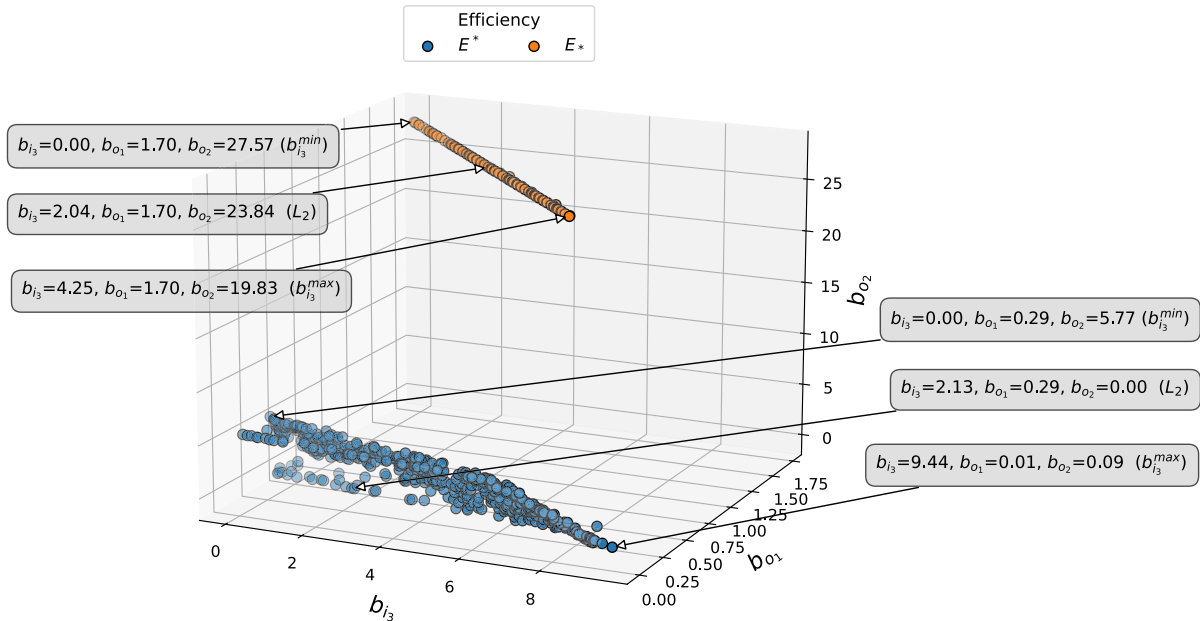
In the case of target $T(E_{KAT,*} \geq 0.95)$, the choice of the best solution is trivial since the entire population converged to a single point. This means that only one performance modification vector allows KAT to attain the same efficiency score under the most unfavorable conditions. Moreover, the necessary achievement of the same efficiency score requires more significant improvements on all inputs than possible achievement (see Fig. 3). Note that the traditional DEA models do not allow flexibility offered by our method in terms of setting both the numerical value of the target efficiency score other than one and its desired level of robustness (necessary or possible).

Let us now focus on POZ, which achieved the highest efficiency scores among the inefficient airports. The analysis involving only its

Table 5

The list of considered targets related to the efficiency scores.

Aim	Necessary vs. possible		Progressive targets				
Airport	KAT	POZ	POZ	KAT	KAT	KAT	IEG
Targets	$E^* \geq 0.95$	$E^* \geq 1.00$	$E^* \geq 0.85$	$E^* \geq 0.70$	$E_* \geq 0.35$	$E_* \geq 0.35$	$E_* \geq 0.25$
	$E_* \geq 0.95$	$E_* \geq 1.00$	$E^* \geq 0.90$	$E^* \geq 0.80$	$E_* \geq 0.55$	$E_* \geq 0.55$	$E_* \geq 0.50$
			$E^* \geq 0.95$	$E^* \geq 0.90$	$E_* \geq 0.75$	$E_* \geq 0.75$	$E_* \geq 0.75$
			$E^* \geq 1.00$	$E^* \geq 1.00$	$E_* \geq 0.95$	$E_* \geq 0.95$	$E_* \geq 1.00$
Factors	i_1, i_2, i_3	i_3, o_1, o_2	i_1, i_2, i_3	o_1, o_2	i_1, i_2, i_3	o_1, o_2	i_3, o_1, o_2
Selection	L -norms	L -norms	CONE_{ASF}	–	–	–	–
Comparison	–	–	CCR	CCR	–	–	–

**Fig. 3.** The solutions representing minimal improvements on inputs that allow KAT to achieve an efficiency score of at least 0.95 possibly and necessarily.**Fig. 4.** The solutions representing minimal improvements on i_3 and outputs that allow POZ to achieve an efficiency score of at least one possibly and necessarily.

input performances indicated that POZ could possibly become efficient. However, even the maximal admissible improvements of inputs would not allow POZ to attain a minimum efficiency score higher than 0.75.

In this case, it would be interesting to investigate whether the increased production of outputs would improve this score.

In Fig. 4, we present the performance modifications that would allow POZ to achieve the minimum and maximum efficiency score

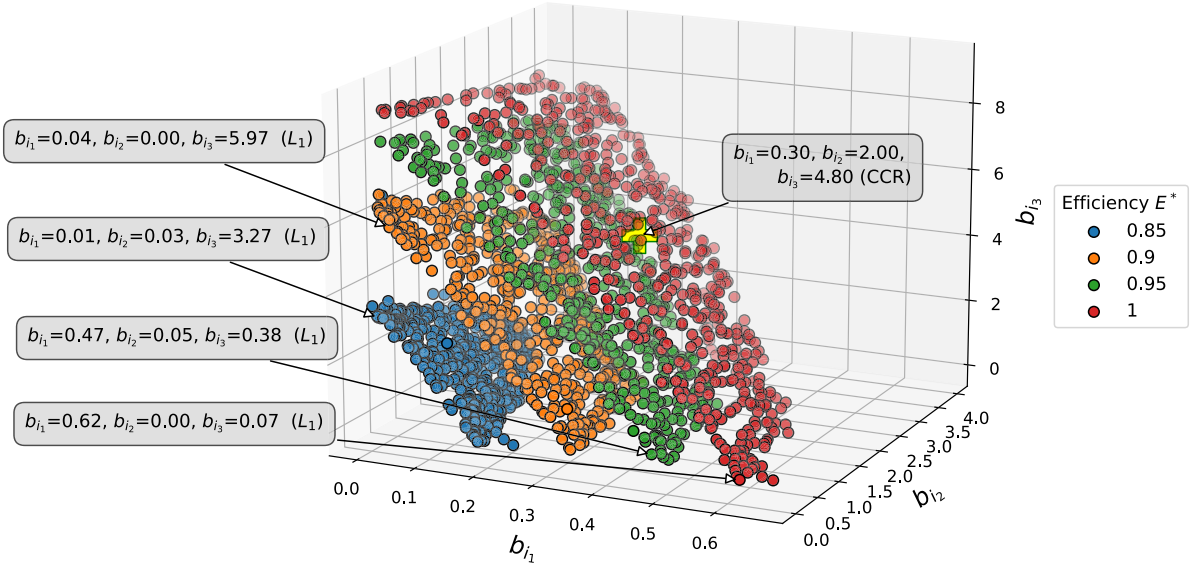


Fig. 5. The solutions representing minimal input improvements that allow POZ to achieve different efficiency score milestones possibly and the solution (marked with a cross) obtained with the standard input-oriented CCR model.

of one. First, we examine what improvements make POZ efficient for at least one weight vector. Let us remind that POZ has achieved the maximum efficiency score of 0.8, so the performance gap that needs to be covered is relatively small. For example, the solution $b_{i_3}^{min}$ indicates that no improvement on i_3 is required to achieve the target. However, the airport would need to serve about 290,000 additional passengers (o_1) per year and increase the number of aircraft movements (o_2) by 5770, which is quite an effort. Therefore, it is more reasonable to choose another solution, e.g., one that minimizes the L_2 norm. Although it requires similar improvement on o_1 , there is no need to serve additional planes if the apron capacity (i_3) was reduced by 2.13. The solution denoted as $b_{i_3}^{max}$ indicates that further improvement on i_3 would allow POZ to reach efficiency with marginal improvements on outputs.

When we implement any of the above solutions, we will make POZ efficient only under particular conditions. To find the required improvements that would guarantee attaining the same efficiency level in the worst-case scenario, we consider target $T(E_{POZ,*} \geq 1)$. The respective minimal performance modifications are presented in Fig. 4. Comparing them with the previous target emphasizes the effort needed to be efficient under all circumstances. First, the number of passengers served per year must be increased by 1.7 million. Since POZ serves only 1.3 million passengers per year, it would be very challenging. Moreover, the annual number of aircraft movements needs to be increased by at least 19,830 provided that the apron capacity (i_3) is also reduced by 4.25. Interestingly, the target can be attained without improving i_3 , but it would require further improvement on o_2 . One may also find a reasonable trade-off between these extreme solutions by choosing the modification vector $[b_{i_3}, b_{o_1}, b_{o_2}] = [2.12, 1.7, 23.71]$ which minimizes the L_2 norm.

4.2.2. The attainment of progressive targets

In this section, we examine how the required improvements increase when we expect a gradual improvement in efficiency. First, we focus on the modifications of inputs that would allow POZ to improve its efficiency score under the most favorable conditions. We divided the performance gap POZ should cover to become efficient into four equally distant steps and specified the appropriate targets. The solutions representing the optimal ways to achieve them are presented in Fig. 5.

On the one hand, the analysis of the extreme solutions revealed that POZ does not need to improve on all factors simultaneously. Each target can be satisfied by reducing the consumption of any pair of inputs. For example, if one decides that it is necessary to maintain the current capacity of a terminal (i_1), POZ can still become efficient by reducing the maximum runway capacity (i_2) by at least 0.3 and the dynamic apron capacity (i_3) by 9.06. In turn, the achievement of the least demanding target $T(E_{POZ}^* \geq 0.85)$ is possible with the significantly lower reduction of apron capacity ($b_{i_3} = 3.35$), whereas the required improvement on i_2 remains at a similar level of 0.28.

On the other hand, when there is no additional preference, it seems reasonable to recommend some well-balanced solution, e.g., one that requires the reduction of i_1 , i_2 and i_3 by 0.31, 1.93, and 4.73 respectively. Besides, if the above improvements could not be implemented in one step, we may propose several intermediate solutions allowing for gradual improvement, e.g., those marked in Fig. 5. Each of them is optimal in terms of the Euclidean norm and leads to achieving a significant efficiency milestone.

When we present the results to the DMs, they might prefer a different solution. In this case, we should employ some MCDA method incorporating DM's preferences and narrowing down the number of possible choices to a small subset of the most satisfying solutions. For this purpose, we decided to use an interactive method called $CONE_{ASF}$ (Kadziński & Słowiński, 2012; Kadziński, Tomczyk, & Słowiński, 2020). This method's preference information comprises a reference point and interactively provided pairwise comparisons. The approach represents such preferences with a compatible Achievement Scalarizing Functions (ASFs), which form a preference cone in the objective space. This cone is systematically contracted with each new pairwise comparison.

Let us provide an example in which $CONE_{ASF}$ is used to support the DM in choosing the most satisfying solution from the set representing minimal modifications that allow POZ to become efficient (see Fig. 5). Initially, the Pareto front contained 526 unique solutions, and the reference point is situated in the origin of the objective space $[0.0, 0.0, 0.0]$. We demonstrate the progress of four iterations, with the more preferred solution in each of them being marked with a red dot and the less preferred solution represented with an orange dot in Fig. 5.

In the first iteration, the DM states that solution $[0.2, 0.04, 8.18]$ is better than solution $[0.49, 0.01, 2.53]$. Note that the preferred solution is evaluated better on the first objective (i.e., it requires less

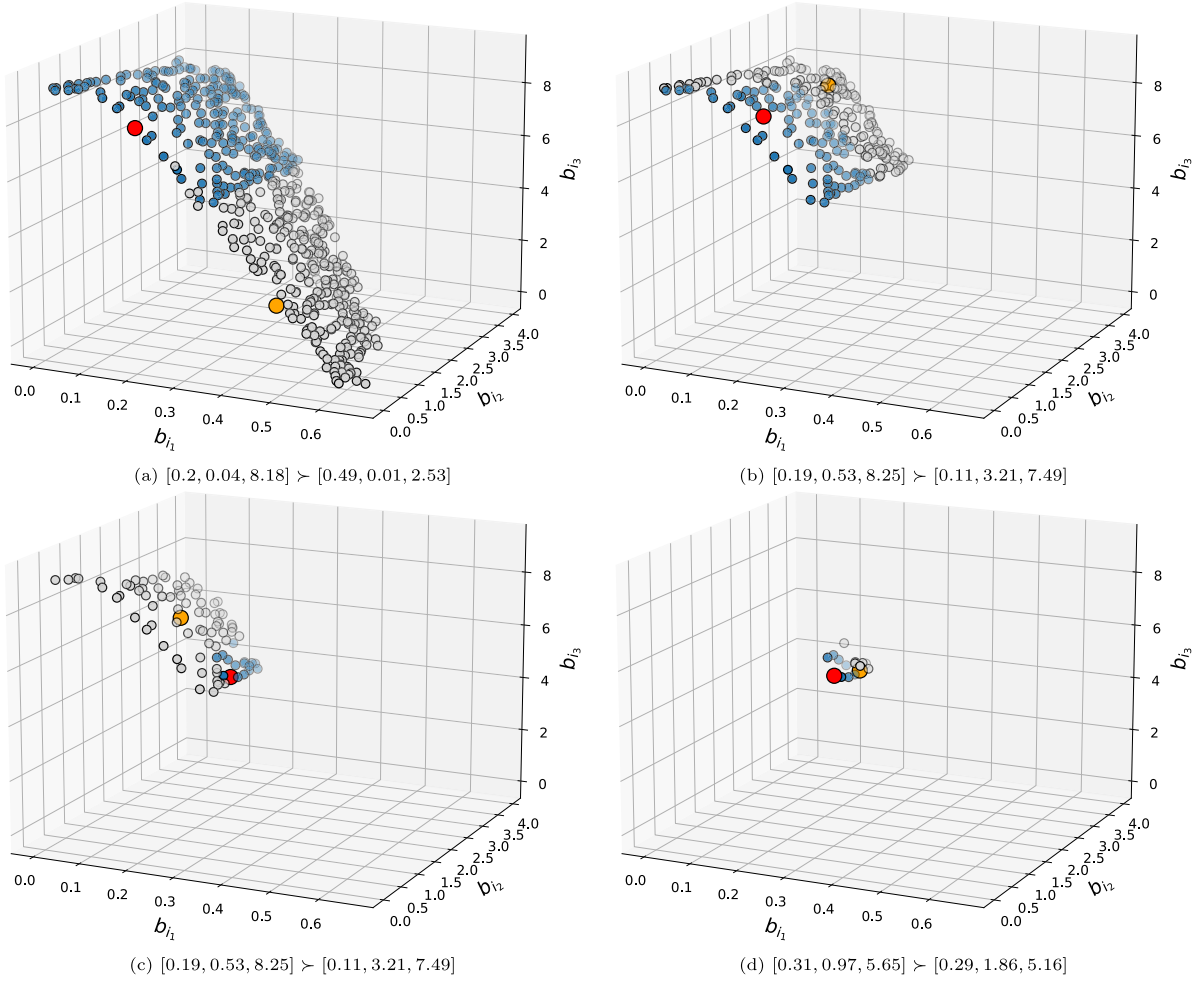


Fig. 6. The sets of solutions after each iteration of $CONE_{ASF}$.

Table 6

The set of eight solutions obtained after the last iteration of $CONE_{ASF}$ method.

No.	b_{i1}	b_{i2}	b_{i3}	No.	b_{i1}	b_{i2}	b_{i3}	No.	b_{i1}	b_{i2}	b_{i3}	No.	b_{i1}	b_{i2}	b_{i3}
1	0.27	1.55	5.74	3	0.26	1.48	5.95	5	0.31	0.97	5.65	7	0.28	1.63	5.50
2	0.30	1.38	5.29	4	0.30	1.56	5.25	6	0.27	1.29	6.00	8	0.30	1.22	5.41

improvement on i_1), slightly worse on the second objective, and significantly worse on the third objective, which means that the DM aims to minimize the necessary improvement on i_1 at the cost of a greater improvement on i_3 . Consequently, based on this pairwise comparison, the method separated the objective space so that the 228 solutions requiring less improvement on i_1 are contained in the preference cone. The solutions that pass to the next iteration are visualized in Fig. 6(a) with blue dots. In turn, the solutions outside the cone, eliminated in this iteration, are outlined in gray.

Then, the DM provides a second comparison: $[0.19, 0.53, 8.25] > [0.11, 3.21, 7.49]$, favoring a more negligible modification on i_2 at the expense of the other inputs. As a result, the cone is narrowed down to contain 91 solutions. The DM selects the next pair among them, stating that $[0.3, 1.22, 5.41] > [0.2, 1.21, 7.37]$. After the third comparison, the number of considered solutions decreased to 16, about 3% of the initial set. At this stage, the remaining solutions are very similar (see Fig. 6(c)). However, it is still difficult for the DM to choose the most favorable one, so she provides another comparison: $[0.31, 0.97, 5.65] > [0.29, 1.86, 5.16]$. Then, the contracted cone contains only eight solutions presented in Table 6, among which the DM chooses $[0.30, 1.22, 5.41]$ as the most satisfactory.

Fig. 5 also depicts the solution obtained using the standard input-oriented CCR model. It allows POZ to achieve efficiency (i.e., a score of one). This solution corresponds to the decrease of the three inputs by $[0.30, 2.00, 4.80]$, i.e., 20% of the airport's current performances. It results from the projection of POZ on the following conical combination of other airports: $0.04832 \cdot WAW + 0.07287 \cdot KRK + 0.20217 \cdot WRO + 0.16319 \cdot GDN$. This solution is Pareto optimal and was also obtained with our method. However, the analysis of Fig. 5 demonstrates the latter's advantage as it computes various performance modifications. This lets the DMs analyze the trade-offs between changes required on different inputs and select among them the one they would be willing to implement in practice.

Let us now focus on output-oriented improvements. The example results obtained for KAT are presented in Fig. 7. In addition to the primary goal of attaining an efficiency score of one for at least one weight vector, we identified the performance modifications that are sufficient to achieve less demanding targets. What needs further explanation is the gradient that fills the background of the chart. It was derived by sampling the space of admissible performance improvements on outputs and represents the maximum efficiency score that KAT would achieve after applying these modifications. Moreover, the

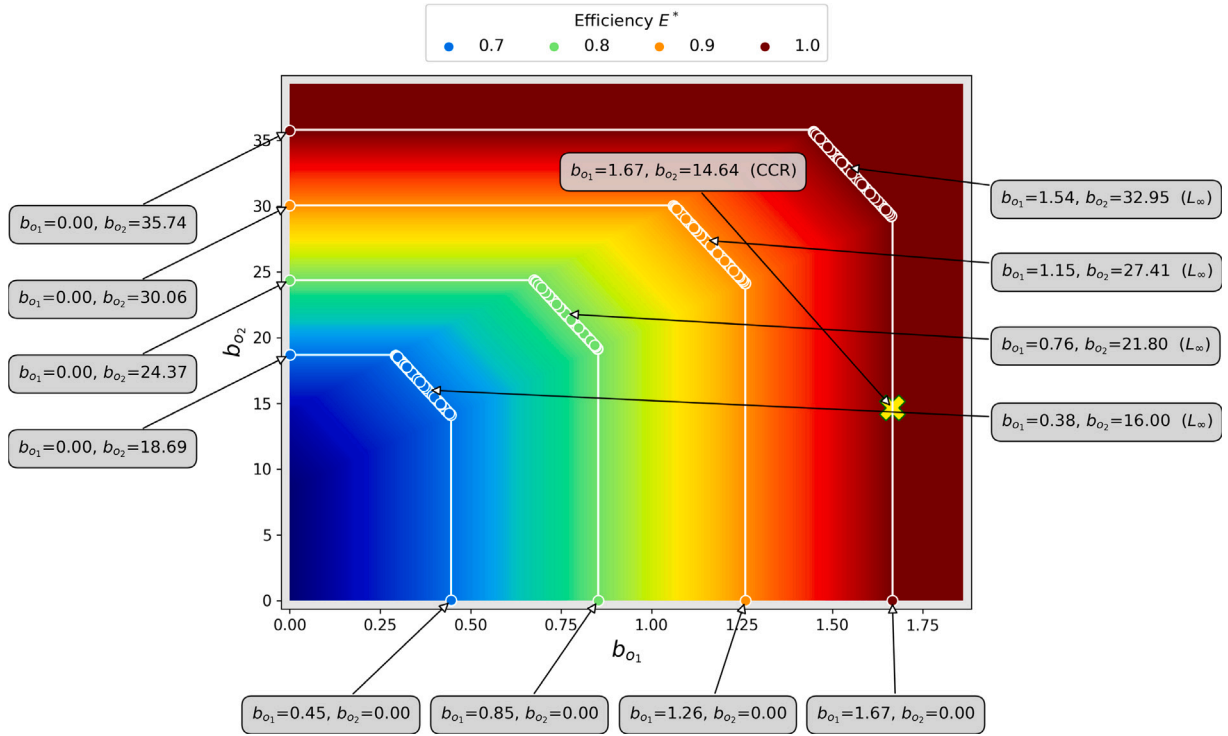


Fig. 7. The solutions representing minimal improvements on outputs that allow KAT to achieve different efficiency score milestones possibly and the solution (marked with a cross) obtained with the standard output-oriented CCR model.

white isoquants indicate the performance modifications that lead to achieving efficiency scores equal to the values specified in the targets. These additional elements prove that the solutions obtained by the proposed method are correct and that the effectiveness of the applied MOO models is high.

Let us discuss the performance changes required to improve KAT's efficiency score, which currently equals 0.59. No matter which target we choose, it can be achieved by improving one output. For example, to attain an efficiency score of 0.8, KAT needs to increase the number of passengers served per year by 850,000 or increase the number of movements by 24,370. To become efficient, KAT would need to serve 35,740 additional planes or 1.67 million passengers. However, these two factors are somehow correlated because serving so many additional passengers involves additional aircraft movements. In this case, one would probably prefer a solution that balances the required modifications. For example, the maximum efficiency score of 0.8 can be achieved by applying the following improvements: $[b_{o1}, b_{o2}] = [0.76, 21.80]$.

Fig. 7 also depicts the solution obtained for KAT with the standard output-oriented CCR model. It postulates the increase of the two outputs by $[1.666, 14.465]$, i.e., by 69.4% of the current outputs of KAT. This solution is only weakly Pareto optimal and, therefore, would not be discovered with our approach. Indeed, we indicate that KAT can be efficient when improving o_1 by 1.666 and leaving o_2 at the already attained level. Thus, the traditional CCR model indicates an excessive modification of the analyzed airport on some outputs, which can be avoided due to the non-radial character of our approach.

The analyses conducted so far indicate that there are various strategies to increase the maximum efficiency score. Such flexibility allows us to choose a balanced solution that requires a reasonable improvement of all factors or focus on a smaller subset of inputs and outputs. Conversely, when our goal is to improve efficiency in the most unfavorable conditions, we must accept the need for significant improvements, and the diversity of available solutions is limited. The analysis of the input-oriented improvements revealed that there is only one optimal way to

increase KAT's minimum efficiency to 0.95 (see Fig. 3). Because this target was ambitious, we performed additional experiments to examine the performance modifications sufficient to achieve lower scores and compare them with the required output improvements. The minimal improvements that allow attaining each target are presented in Fig. 8. In each case, the entire population of solutions converged to one point. This makes sense because the efficiency score is calculated as a ratio of a weighted sum of outputs to a weighted sum of inputs. For example, let us assume that KAT is expected to achieve the minimum efficiency score of 0.55, and reducing consumed inputs is impossible. Then, it is necessary to increase the production of o_1 and o_2 by at least 2.24 and 42.31. If one of these improvements cannot be fully implemented, the most unfavorable weight vector will emphasize the shortage of that output, thus preventing KAT from reaching the target.

Finally, let us focus on IEG, which is the least efficient airport in the dataset. IEG has a higher apron capacity than most airports, which means it can serve many planes per hour. However, the output performances indicate that the number of passengers and planes that IEG actually serves is lower by orders of magnitude when compared to the other airports. The combination of these factors resulted in a low efficiency score, which in the worst-case scenario equals 0.001. To examine the performance improvements required to improve this score, we have specified four targets concerning different efficiency thresholds. The minimal changes that allow achieving these targets are presented in Fig. 9. Since the performance on i_1 cannot be improved (IEG has the lowest capacity of a terminal among the airports) and the runway capacity (i_2) is already relatively low, the analysis only involves modifying i_3 and both outputs. Note that such a mix of inputs and outputs cannot be considered with the standard CCR model, which admits the modification of either inputs or outputs. Moreover, our framework works with any arbitrarily indicated subset of factors, e.g., involving a proper subset of inputs or outputs. On the contrary, the CCR model simultaneously modifies all factors of a given type.

IEG can achieve each target without any improvement on i_3 . However, in this case, it is necessary to increase the production of both

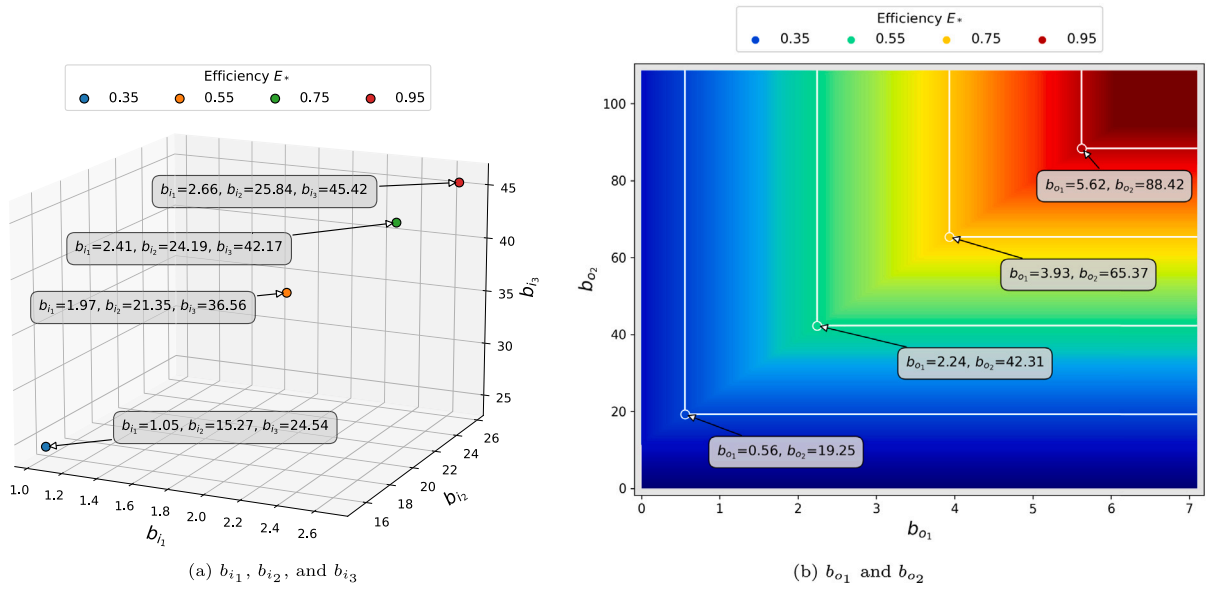


Fig. 8. The solutions representing minimal improvements that allow KAT to achieve different efficiency score milestones necessarily.

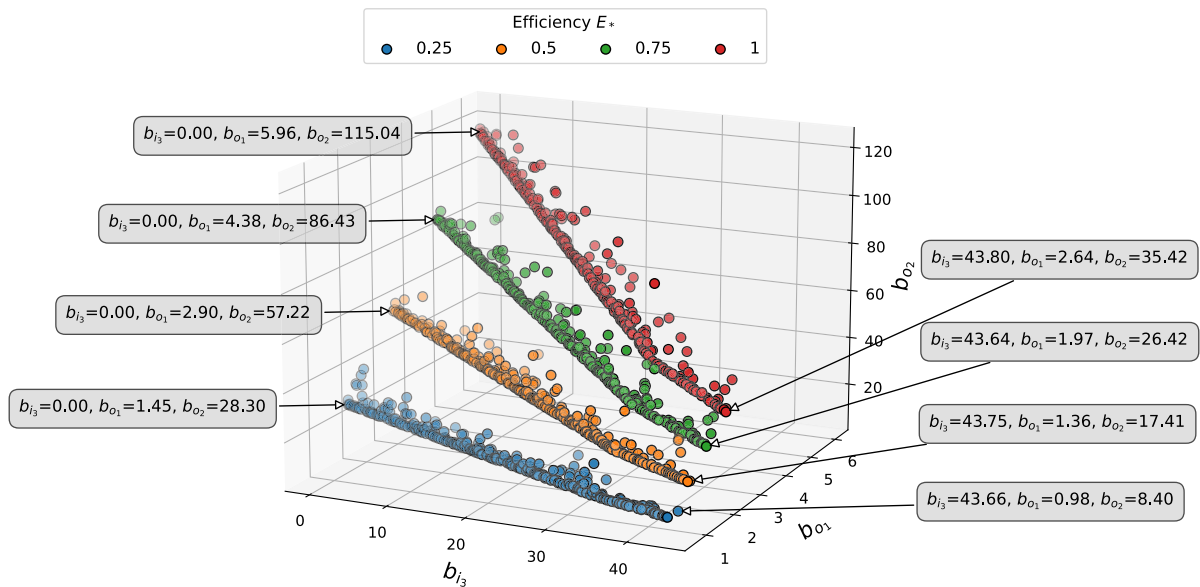


Fig. 9. The solutions representing minimal improvements on i_3 and outputs that allow IEG to achieve different efficiency score milestones necessarily.

outputs significantly. For example, to attain the minimum efficiency score of 0.5, IEG needs to increase the passenger traffic by at least 2.9 million and the number of aircraft movements by 57,220. Nevertheless, in contrast to the previous example where inputs and outputs were considered separately, it is possible to choose an alternative solution that would allow compensating the necessary improvements on outputs by reducing the excessive use of i_3 . If we choose the solution that requires reducing the apron capacity (i_3) by 43.75, the necessary improvements on o_1 and o_2 will be reduced to 1.36 million passengers and 17.41 planes, respectively. The analysis of the remaining targets shows that the respective extreme solutions exhibit similar dependencies. The necessary improvements on o_1 and o_2 are about two and three times higher when i_3 is not changed than in the scenario where the improvement on i_3 is maximal.

In the eAppendix, we discuss the results for several targets referring to efficiency ranks and preference relations. Also, we present the

outcomes for some example targets to be achieved while modifying the performances of other considered units within some reasonable bounds. This makes the target attainment even more challenging, implying the need for more significant modifications of inputs and outputs.

5. Conclusions

Our primary focus was on the improvement scenarios indicating the performance modifications required to attain some pre-defined target. The traditional efficiency analysis considers only the improvements needed to attain an ideal efficiency score. In turn, we referred to the improvements required to achieve a pre-defined score, rank, or the truth of a preference relation. Moreover, we accounted for two settings under which these targets can be attained necessarily or possibly.

For all scenarios, we formulated Multiple Objective Optimization problems, minimizing the required modifications of inputs and/or outputs needed to reach the pre-defined aim. The proposed framework leaves excellent flexibility to the analyst, who can indicate the factors that should be modified (e.g., by mixing inputs and outputs) and define constraints on the feasible performance modifications. Although the formulated problems can be solved using various optimization algorithms, we opted for an evolutionary approach. Specifically, we applied a suitably adjusted variant of the NSGA-II method.

Our method contributes to developing robustness and sensitivity analyses in DEA. On the one hand, we build on the robust approach, which accounts for the uncertainty related to the selection of input and output weights already at the stage of deriving the recommendation for a particular problem. In particular, we deem some conclusions robust if valid for all feasible weights. On the other hand, we contribute to the sensitivity analysis by verifying how the already derived recommendation would be affected under alternative assumptions on the DMUs' performances. However, this is conducted in a specific setting oriented toward modifying the performances minimally so that a desired outcome is attained.

The use of the proposed framework was illustrated in the problem of analyzing the efficiency of Polish airports. We first discussed the outcomes of robustness analysis in terms of ranks, scores, and relations. These served as input to formulate some performance targets. Then, we focused on the underperforming airports and demonstrated how they could attain better scores or ranks or outrank the more favorable units. When discussing the results, we aimed to illustrate the capabilities offered by the introduced methodology. Therefore, we showed the differences between attaining more and less demanding targets (e.g., attaining the first rank is more demanding than the second position, whereas the necessary targets that need to hold for all feasible weights are more demanding than the possible ones that are validated already with at least one weight vector). Moreover, we demonstrated how the framework could be used to modify either inputs or outputs, as done in the traditional DEA model, or both types of factors simultaneously.

The method presented in this paper can be extended in numerous ways. First, we plan to apply the proposed approach to other efficiency models. In the current work, we focused on ratio-based efficiency, which is the most widely used in Data Envelopment Analysis. However, other models, such as an additive or value-based efficiency method, can be incorporated into the proposed framework. Second, we will propose strategies for step-wise benchmarking (Lozano & Calzada-Infante, 2018). In some cases, the required performance modifications may be too significant to attain in a single step. Then, it might be desirable to design a long-term strategy for attaining the targets in a few steps. Finally, we envisage the application of the proposed method in real-world case studies. The most appealing application fields are transport, energy, and education.

CRedit authorship contribution statement

Dariusz Grynia: Conceptualization, Methodology, Software, Investigation, Data curation, Writing – original draft, Visualization. **Miłosz Kadziński:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Supervision, Project administration, Funding acquisition. **Mladen Stamenković:** Conceptualization, Methodology, Validation, Writing – revision, Supervision.

Data availability

Data is available as part of the paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cie.2024.110065>.

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